Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
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Practice Exam — Week 11: Diagnostics, Model Selection & Long Memory

Duration: 60 minutes · No calculator · Answer on paper

Name: ____________________ Surname: ____________________

SCIPER (Student Card Number): ____________________

Exercise	**Points**
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2.

Problem 1 (5 points)

(Residual diagnostics and the danger of R^2.) A correctly specified, stationary AR(1) model $X_t = \phi X_{t-1} + \varepsilon_t$ ($|\phi| < 1$) is fitted with the true parameters known. Recall that the one-step fitted value is $\hat{X}_t = \phi X_{t-1}$, the residual is $e_t = X_t - \hat{X}_t$, and that for white-noise residuals each sample residual autocorrelation $\hat{r}_\tau$ is approximately $\mathcal{N}(0, 1/n)$.

(a) Show that the residual equals the innovation, $e_t = \varepsilon_t$, and use this to state which of the four standard residual checks (constant zero mean, constant variance, no autocorrelation, Gaussianity) the residuals must pass and why. (1.5 pts)

(b) For $n = 400$ observations, give the numerical pointwise 95% correlogram band $(-1.96/\sqrt{n}, 1.96/\sqrt{n})$ for the residual ACF, and explain in one sentence why eyeballing this band lag-by-lag is *not* a substitute for a portmanteau test. (1.5 pts)

(c) Show that for an AR(1) the population $R^2 = 1 - s_e^2/s_y^2$ equals ϕ^2, where s_e^2 is the residual variance and $s_y^2 = \mathrm{Var}(X_t)$. Evaluate it at $\phi = 0.6$, and explain why a "low" value here is no indictment of the model and why a higher R^2 obtained by fitting an AR(4) to the same data is no endorsement. (2 pts)

Problem 2 (5 points)

(Information criteria: AIC, AICc, BIC.) For a model with k estimated parameters fitted to n observations with maximised likelihood L,

$$\mathrm{AIC} = -2\ln L + 2k, \qquad \mathrm{AICc} = -2\ln L + 2k\,\frac{n}{n-k-1}, \qquad \mathrm{BIC} = -2\ln L + k\ln n,$$

and for an ARMA(p, q) the parameter count is $k = p + q + 1$. Smaller is better for all three.

(a) Prove the identity

$$\mathrm{AICc} - \mathrm{AIC} = \frac{2k(k+1)}{n-k-1},$$

and state precisely the conditions on n and k under which AIC and AICc become approximately equivalent. (1.5 pts)

(b) Consider a mean-zero ARMA(p, q) model in which exactly m of the $p + q$ AR and MA coefficients are *known in advance and fixed to zero* (only the remaining ones, and the innovation variance, are estimated). Give the estimated-parameter count k and write down the resulting AICc in closed form. (1.5 pts)

(c) A full AR(4) model ($k = p+q+1 = 5$) is compared with a constrained AR(4) in which $m = 2$ coefficients are fixed to zero, on $n = 30$ observations. Compute the AICc complexity penalty $2k\,n/(n-k-1)$ for each model, and state by how much the constrained model's $-2\ln L$ is allowed to *exceed* the full model's before AICc would prefer the full model. (2 pts)

Problem 3 (5 points)

(Portmanteau tests and the degrees of freedom.) For $n = 200$ observations a mean-zero ARMA$(2,1)$ model is fitted, and the residual sample autocorrelations at lags 1 to 5 are

$$\hat{r}_1 = 0.05, \quad \hat{r}_2 = -0.08, \quad \hat{r}_3 = 0.12, \quad \hat{r}_4 = 0.04, \quad \hat{r}_5 = -0.10.$$

Recall the Box–Pierce statistic $Q_m^{\mathrm{BP}} = n \sum_{\tau=1}^m \hat{r}_\tau^2$ and the Ljung–Box statistic $Q_m = n(n+2)\sum_{\tau=1}^m \hat{r}_\tau^2/(n-\tau)$, both approximately χ^2_{m-p-q} under the null that the residuals are white up to lag m. Useful upper-5% quantiles: $\chi^2_{2,0.95} = 5.991$, $\chi^2_{3,0.95} = 7.815$, $\chi^2_{4,0.95} = 9.488$.

(a) Compute the Ljung–Box statistic Q_5 (show the term-by-term sum). (2 pts)

(b) Test $H_0 : \rho_1 = \cdots = \rho_5 = 0$ at the 5% level for the ARMA$(2,1)$ fit: state the degrees of freedom, the critical value, and the conclusion. (1.5 pts)

(c) Suppose the *same* residual autocorrelations had instead come from an AR(1) fit. Redo the test (state the new df, critical value, conclusion) and explain, in one sentence, why getting the degrees of freedom right (df $= m - p - q$, not m) is essential. (1.5 pts)

Problem 4 (5 points)

(Revision of Week 10: forecasting a Gaussian AR(1).) Let $X_t = \phi X_{t-1} + \varepsilon_t$ be a causal Gaussian AR(1) with $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, and recall that for a causal ARMA process the h-step forecast error variance is $\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$, where $\{\psi_j\}$ are the MA(∞) weights.

(a) Give the MA(∞) weights ψ_j of the AR(1) and, using that the best linear predictor zeroes future innovations, show that $X_{n+h}^n = \phi^h X_n$. (1.5 pts)

(b) Show that $\text{Var}(e_n(h)) = \sigma^2 \dfrac{1 - \phi^{2h}}{1 - \phi^2}$ and verify that, as $h \to \infty$, both the point forecast and the error variance converge to their stationary limits (the mean 0 and the marginal variance γ_0 respectively); identify γ_0. (1.5 pts)

(c) Take $\phi = 0.6$, $\sigma^2 = 4$ and the last observation $X_n = 5$. Compute the two-step forecast X_{n+2}^n, its forecast-error variance, and a 95% Gaussian prediction interval for X_{n+2} (use $z_{0.975} = 1.96$, $\sqrt{5.44} \approx 2.332$). (2 pts)